

Mycolris

A proof-of-concept exploring fungal colonial morphology as a polymorphic analogue of the human iris for physical unclonable functions for biometric classification

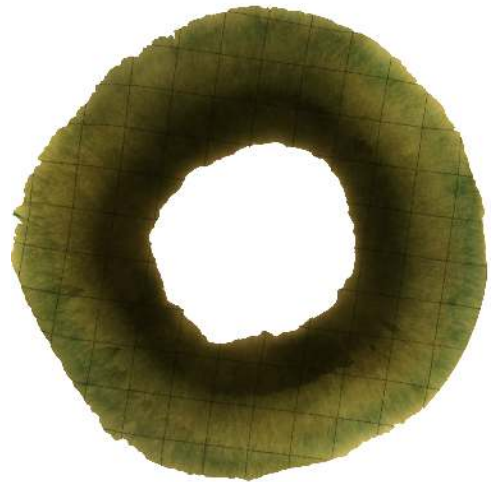
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Fungal colony morphology as an analogue to the human iris

This concept proposes that the macroscopic morphology of a fungal colony can serve as a functional analogue to the human iris within biometric and physical verification systems, and can be processed within the same mathematical and computational regime as iris biometrics. In an AI-saturated context where proof of physicality, personhood, and provenance are increasingly valued, the combination of biological entropy, iris-like biomimicry, and cultural legibility offered by fungal colonies represents a distinctive expression of this broader verification paradigm.

Like the human iris, a fungal colony exhibits complex, radially organized, high-entropy patterning that emerges through self-directed growth rather than explicit design. These patterns are shaped by local growth dynamics, branching behavior, nutrient gradients, and environmental micro-variations, producing structures that are both highly distinctive and difficult to reproduce.

Fungal colonies display concentric zoning, radial fissures, filamentous textures, and stochastic microstructures that closely resemble the spatial complexity leveraged in iris recognition systems. These features are not encoded, printed, or fabricated; they arise through biological growth processes that embed irreducible physical history into the final morphology. Crucially, colonies can be cultivated at a one-to-one physical scale with the human iris, concentrating structural information within the same spatial footprint and resolution range used in iris scanning. As a result, each colony constitutes a naturally generated, physically embodied pattern suitable for use as a unique identifier, containing readable information at the same scale and geometry as established iris biometrics.



Polymorphic entropy and orthogonal feature spaces

Unlike the human iris, which derives most of its discriminative power from texture within a narrow anatomical region, fungal colonies provide access to multiple orthogonal dimensions of entropy within a single physical object. Texture-based features analogous to iris crypts, furrows,

and radial striations can be extracted from grayscale or contrast-enhanced images of the colony interior. These features alone are sufficient to support strong separability between individual specimens.

Beyond texture, fungal colonies can incorporate additional, independent entropy sources. Certain species produce rare, chemically complex pigments that are difficult to synthesize or precisely reproduce, such as xylindein. The chemical complexity and limited synthetic accessibility of these pigments act as a natural barrier to counterfeiting, while also introducing spectroscopic and chromatic signatures that can be captured through multispectral or color-aware imaging, adding a chemical dimension of uniqueness that is fundamentally orthogonal to morphology. Because pigment expression is influenced by metabolic state and growth history, it further deepens the irreversibility of the pattern.

Finally, the outer boundary of a colony encodes growth kinetics and spatial negotiation with its environment. Colony perimeter shape, edge roughness, and asymmetry provide a third, complementary feature space that can be used independently or fused with internal texture and pigment information. Together, these dimensions form a rich, polymorphic entropy landscape in which multiple independent features can be combined, weighted, or selectively sampled depending on application requirements.

Comparison with the human iris: privacy and revocability

While the human iris is among the most stable and information-dense biological identifiers, it is inseparably bound to the individual and therefore tightly coupled to personal identity, bodily autonomy, and long-term privacy risk. Iris patterns cannot be meaningfully revoked; once compromised, they remain compromised for life. This creates structural challenges for large-scale deployment, regulation, and ethical governance.

In contrast, fungal colony identifiers are non-personal and externalized. They are not derived from the human body, do not encode personal biological information, and are not permanently linked to an individual. A compromised or retired colony can simply be destroyed and replaced with a newly grown instance, restoring security without long-term consequence. This inherent revocability allows fungal-based identifiers to function more like cryptographic keys than biometric traits, while retaining the physical unclonability advantages of biological systems.

Because the identifier exists as a separate physical artifact, ownership, custody, and consent can be managed independently of the person, enabling verification systems that avoid many of the ethical and regulatory pitfalls associated with biometric surveillance.

Proof-of-Concept

This proof-of-concept study demonstrates that fungal colony morphologies can function as effective morphological analogues to the human iris within a standard iris-analysis workflow. By cultivating, stabilizing, and optically imaging pigmented fungal colonies, the work shows that

their inherently radial, high-entropy growth patterns can be segmented, normalized, and encoded using computational steps directly parallel to those employed in human iris biometrics, including annular isolation, polar unwrapping, feature extraction, and Hamming-distance-based comparison. The resulting encodings are shown to be both highly reproducible for repeated measurements of the same colony and strongly discriminative across different colonies, establishing fungal growth as a viable biological substrate for iris-style pattern analysis.

Cultivation and Stabilization

Culture preparation

Colonies were cultivated in 60 mm sterile polystyrene Petri dishes containing malt-extract agar (MEA) overlaid with cellulose-nitrate membranes (Whatman-type, 0.45 μm). A homogeneous inoculum was prepared by dispersing a mature stock culture into sterile distilled water to form a fine hyphal suspension. A 3D-printed PLA inoculation template was used to define the precise position and volume of the inoculum. Each well in the template delineated a circular loading zone at the geometric center of the membrane. A 100 μL aliquot of the cell suspension was pipetted into each well, producing consistent initial conditions across replicates while preserving stochastic variation at the level of hyphal fragment distribution.

This micro-scale randomness in fragment number, orientation, and spatial dispersion constitutes one of the entropy sources for the system. Although the inoculation geometry was fixed, the microscopic heterogeneity of starting material determined the eccentricity and anisotropy of early colonial growth, ensuring that no two colonies shared identical initiation trajectories.

Incubation and colony development

Following inoculation, plates were sealed with Parafilm and incubated at 24 °C in a benchtop incubator for 6–12 weeks. During this period, colonies expanded radially across the cellulose-nitrate substrate, forming concentric and branching zonation patterns characteristic of *C. aeruginascens*. The pigment xylindein developed progressively from pale turquoise to deep blue-green, providing a high-contrast optical field suitable for transmitted-light imaging.

Drying and resin stabilization

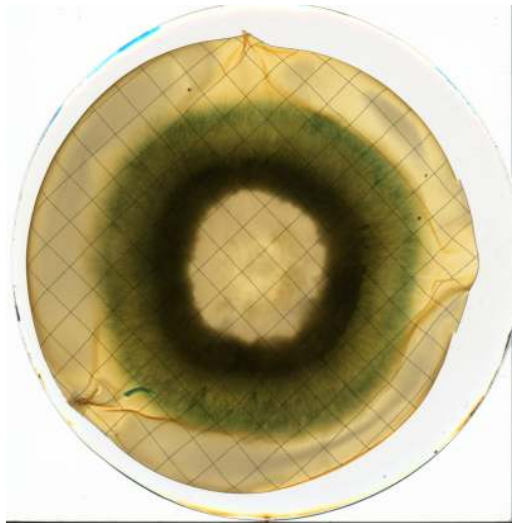
After full pigment expression, the PLA template was carefully removed and each cellulose-nitrate membrane was detached from the agar surface using sterile forceps. Membranes were air-dried at ambient conditions for 48 hours to remove residual moisture and arrest growth.

Each dried membrane, now bearing a complete pigmented colony, was placed in a 50 × 10 mm silicone casting mold and embedded in a UV-protective, non-toxic epoxy resin. To ensure complete infiltration and elimination of trapped air, the molds were vacuum-infused at -30 in Hg for 5 minutes, then released and allowed to cure at room temperature for 72 hours. Upon

demolding, the resulting transparent epoxy pucks encapsulated the cellulose-nitrate membrane and its fungal morphology in a stable, optically clear matrix.

Resulting artifact

The cured disc (approximately 50 mm in diameter and 10 mm thick) constitutes the physical token prototype. Within each resin body, the preserved colony forms a permanent morphological record: a unique, high-entropy growth pattern stabilized for optical interrogation and long-term archival storage. These resin-embedded tokens served as the material basis for subsequent imaging, feature extraction, and distance-metric analyses described in the following sections.



Example of epoxy resin stabilized token prototype (specimen 4).

Optical Imaging

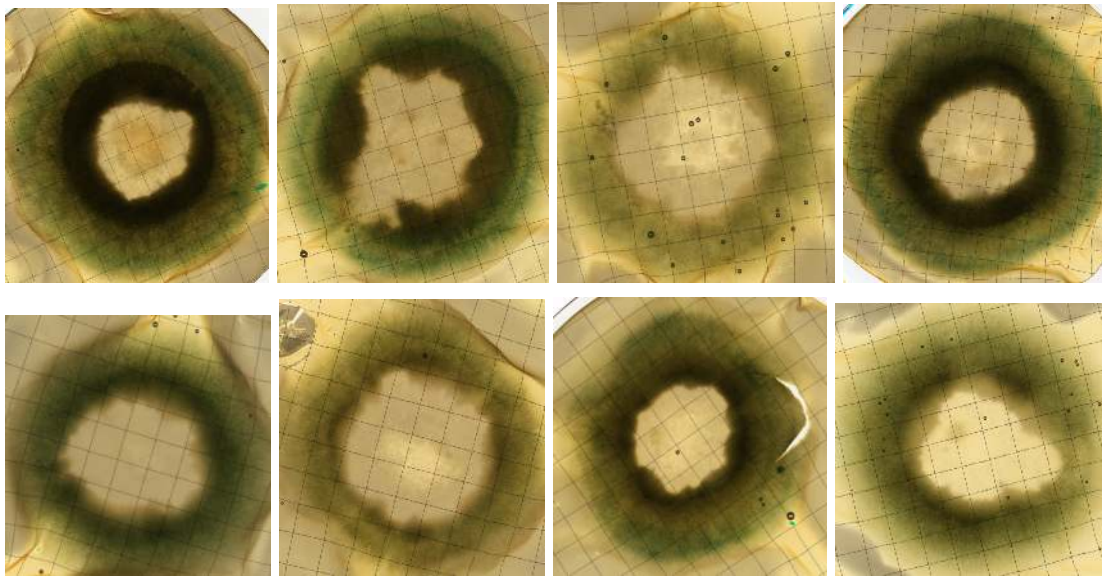
Optical imaging of the stabilized tokens was performed using an Epson Perfection V600 Photo Scanner, selected for its integrated transparency unit and ability to deliver high-resolution, uniformly illuminated transmitted-light scans. Each resin-embedded colony was imaged in photo mode using the Epson Scan 2 software utility under the following standardized parameters:

- **Mode:** Color positive film
- **Color depth:** 24-bit RGB
- **Resolution:** 1600 dpi (High Quality setting)
- **File format:** PNG (lossless compression)
- **Scan area:** 2 × 2 inches (baseline)

Each image captured the full resin disc and surrounding boundary under consistent backlighting, ensuring reproducibility across specimens. Following acquisition, images were cropped manually to the colony's outer perimeter to remove the clear resin margins and isolate the region of biological morphology. The resulting cropped images served as the standardized inputs for subsequent featurization and mycocode generation.

The Epson Perfection V600 was determined to be an optimal imaging instrument for the prototype phase. Its transparency unit enables even, diffuse transmitted-light illumination comparable to dedicated optical or biomedical scanners, while its flatbed configuration minimizes variation in focus and geometry between scans. The combination of high bit depth, fixed optical path, and readily repeatable parameters provides a consistent imaging baseline, essential for verifying intra-token stability and inter-token distinctness in later analyses.

By standardizing these imaging conditions, each specimen could be rendered into a reproducible digital field suitable for quantitative feature extraction, establishing the foundation for the subsequent encoding and Hamming-distance verification workflows.



Eight epoxy-stabilized token prototypes prepared for the initial proof-of-concept. Each specimen is shown under transmitted light using a high-resolution Epson Perfection V600 photo scanner. Despite being cultured under identical conditions, the colonies exhibit pronounced variation in pigmentation, texture, and boundary morphology, demonstrating the intrinsic stochasticity of fungal growth as a source of physical entropy.

MycoCode Generation

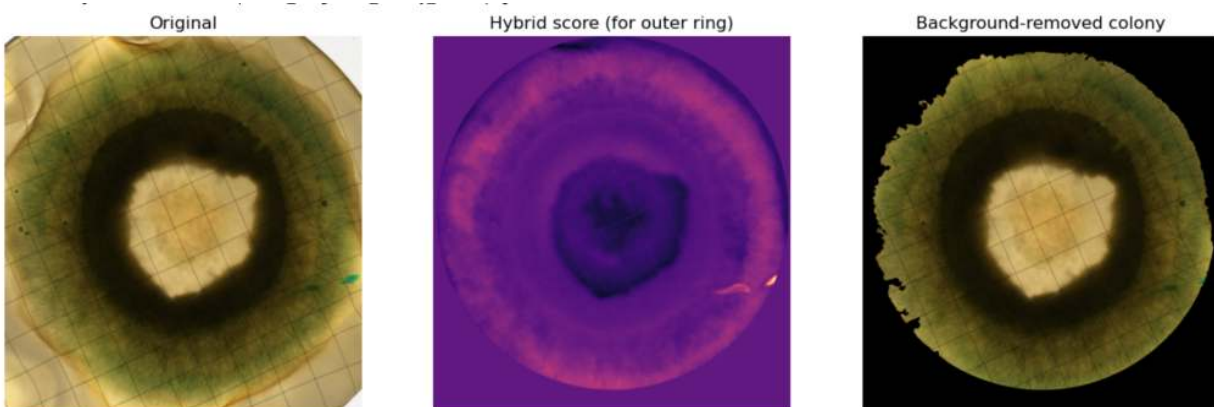
Rationale

A typical iris-scan processing pipeline begins with segmentation, where the circular iris is isolated from the surrounding sclera, eyelid, and pupil using contrast and edge detection. The isolated iris region is then normalized—mathematically “unwrapped” from its circular geometry into a rectangular coordinate map that preserves spatial relationships along radial and angular dimensions. This normalized map is featurized through phase-encoding filters (commonly Gabor or Log-Gabor wavelets) that convert local textural variations into a compact binary representation known as the iris code. Verification occurs by comparing two iris codes via a Hamming distance threshold to determine similarity.

The rationale for emulating this workflow in the present system is to demonstrate that a fungal colony—whose pigmentation and radial morphology naturally mirror the iris—can be processed using the same computational logic that underpins biometric identity. By segmenting the colony, normalizing its radial structure, and encoding its texture into a binary mycocode, the pipeline directly parallels each stage of iris recognition, demonstrating that a non-human, biogenic morphology can produce stable, high-entropy feature maps suitable for identity verification within an analogous biometric framework.

Masking ('Iris' and 'Pupil')

The first stage of the pipeline isolates the fungal colony from its resin background to define the analytical region of interest. Each transmitted-light image was processed in Python using OpenCV and scikit-image libraries. The algorithm first detects the circular resin boundary using a Hough transform or, if necessary, a contour-based fallback. The detected disc is slightly contracted ($\text{DISC_SHRINK} \approx 0.97$) to exclude glare from the resin edge. The image is then converted to CIE Lab* color space, and both the lightness (L^*) and green–magenta (a^*) channels are Gaussian-smoothed ($\sigma = 5$) to emphasize large-scale structure over pixel noise. Local contrast enhancement (CLAHE) on the L^* channel improves separation between the pigmented colony and surrounding substrate. A hybrid score is calculated as a weighted combination of normalized darkness ($-L^*$) and greenness ($-a^*$) values, tuned to highlight xylindein-rich regions ($W_{\text{GREEN}} = 1$, $W_{\text{DARK}} = 0.3$). Adaptive Otsu thresholding of this score yields a binary mask of the colony, which is then morphologically filled to capture the entire growth region. The mask is applied to the RGB image to remove background pixels, producing two standardized outputs—*colony_mask.png* (binary segmentation) and *colony_clean.png* (background-removed image)—that serve as the foundation for all subsequent feature extraction and encoding steps.

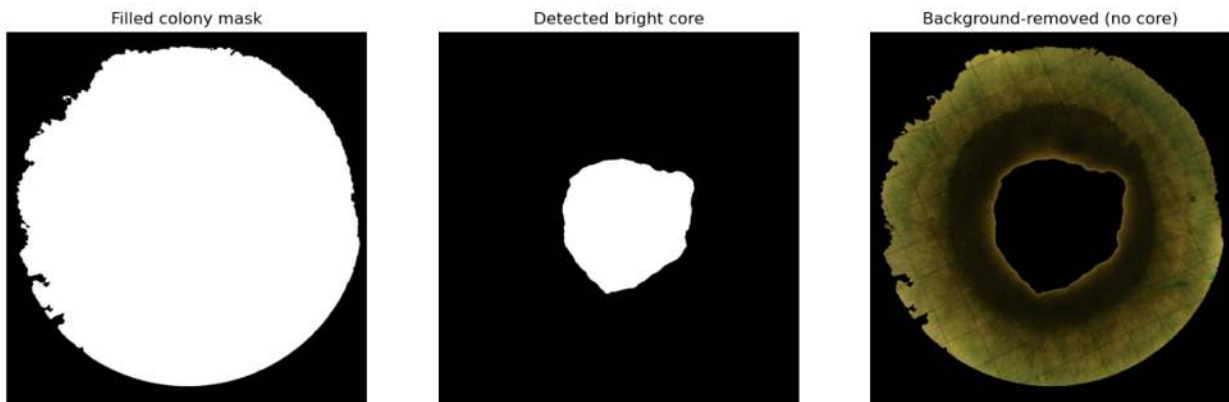


Example of original input image, darkness and greenness scoring to identify colony boundaries, and masked image (specimen 1).

Following initial segmentation, a secondary processing step removes the colony's central bright region ("core") to ensure that downstream texture analysis emphasizes the complex peripheral morphology rather than the saturated or optically uniform center. In the context of the iris

analogy, this central region functions as the pupil—a zone of high optical density or reflectance that often lacks fine texture relative to the annular growth zone. Using the previously defined colony mask and lightness channel (L^*) from the CIE Lab* color space, the algorithm estimates the colony's equivalent radius from its filled area and defines an inner search region encompassing up to 90 % of that radius. Within this region, a brightness threshold—determined adaptively by Otsu's method or optionally by percentile—is applied to detect high-luminance pixels corresponding to the colony's pupil. Small or disconnected bright artifacts are removed, and only the largest central connected component nearest the colony centroid is retained as the pupil mask. Morphological closing and opening operations (disk radius = 7 px) smooth its edges and eliminate small voids.

The refined pupil mask is then subtracted from the full colony mask, yielding a “no-core” binary mask and corresponding background-removed image (*colony_mask_nocore.png*, *colony_clean_nocore.png*). This exclusion isolates the pigment-rich, structurally diverse annular region most analogous to the iris stroma in human biometrics, preparing the image for radial normalization and feature extraction. Notably, the colony pupil often exhibits slight eccentricity or off-center geometry arising from stochastic inoculation and differential pigment deposition—features that may themselves serve as an orthogonal, independent source of morphological entropy distinct from the surrounding annulus.



Example of colony (annular) and core ('pupil') masks and the final masked image (specimen 1), isolating the 'iris' region of the token for unwrapping and MycoCode generation.

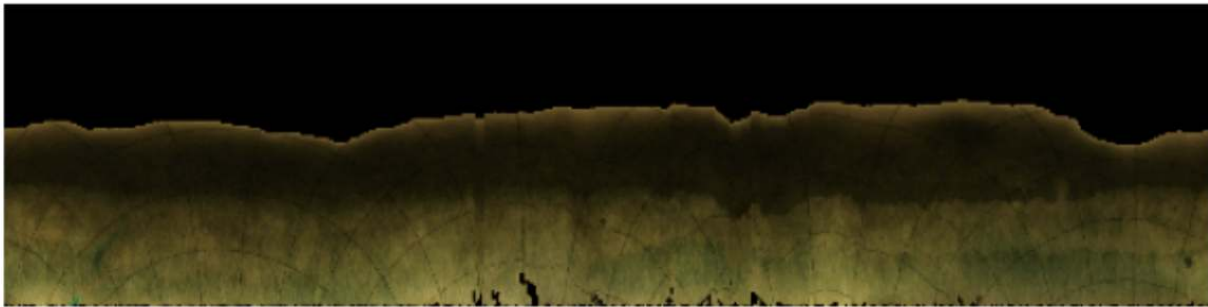
Polar Unwrapping and Boundary-Normalized Sampling

This step converts the segmented colony into a normalized, iris-style “strip” by unwrapping the annulus in polar coordinates and scaling radii per angle to the true colony boundary. The routine first resolves inputs (favoring the core-excluded mask and clean images) and computes the colony centroid from the binary mask. For each of **ANGULAR_BINS** discrete angles, it traces a ray outward from the centroid, measures the mask's per-angle boundary length $L(\theta)$ via a profile scan, replaces degenerate rays with a local median, and optionally applies a circular median smoothing window (**L_SMOOTH_WIN**) to stabilize boundary estimates.

A radial sampling grid of **RADIAL_BINS** rows is then built by linearly spacing r from **INNER_FRAC** (typically 0.0 to include the inner edge of the annulus) to 1.0 and multiplying by the corresponding $L(\theta)$, yielding per-angle, boundary-normalized radii. Using OpenCV's `remap`, the RGB image is sampled with bilinear interpolation (mask with nearest-neighbor) to produce a rectangular unwrapped image whose columns encode angle and rows encode normalized radius, preserving fine texture circumferentially while equalizing radial scale across eccentric boundaries.

The outputs—`<base>_colony_unwrapped_rgb_R{R}_A{A}.png` and the corresponding mask—provide a standardized, rotation-friendly representation analogous to iris “rubber sheet” normalization for downstream feature extraction and code generation.

Unwrapped RGB (R=128, A=512)



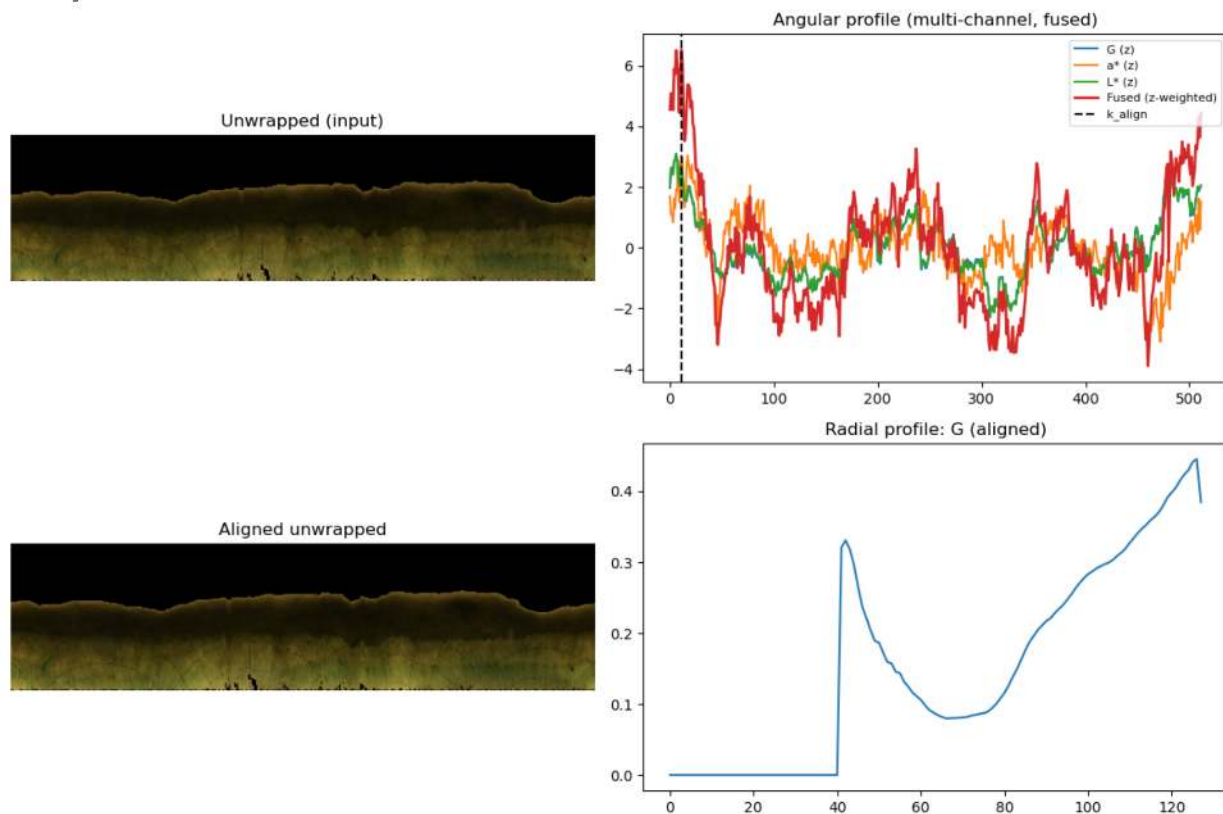
Example of the unwrapped colony “iris” (rubber-sheet normalization, specimen 1). The circular colony has been transformed into a rectangular coordinate map where each column corresponds to angular position and each row to normalized radial distance from the colony center. This representation preserves the colony’s circumferential texture while equalizing radial scale, directly mirroring the normalization step used in conventional human iris recognition workflows.

Angular Canonicalization and Mycocode v0.1 Construction

The final stage of the proof-of-concept pipeline establishes a consistent angular orientation for each unwrapped specimen and extracts its first numerical feature encoding. Because each colony is radially symmetric but lacks a fixed orientation, angular canonicalization ensures that features derived from different specimens (or rescans of the same specimen) are rotationally aligned prior to comparison. To determine a canonical zero-angle, the algorithm computes angular intensity profiles across a mid-radial band (30–80 % of the colony radius) for multiple color channels: green (G), L^* and a^* from CIE $L^*a^*b^*$, and saturation (Sat) from HSV. These per-angle means are standardized (z-scored) and fused into a single orientation profile weighted by chromatic relevance ($G = 1.0$, $a^* = 0.8$, $L^* = 0.6$). The peak of this fused profile defines the reference direction, and the unwrapped image and mask are circularly shifted so that this angle becomes column 0, producing a canonicalized unwrapped representation (`_unwrapped_rgb_aligned.png`).

From this aligned image, the algorithm calculates mask-aware radial profiles for five spectral–structural features—G, a^* , b^* , saturation, and L^* —averaging pixel values along each

radius. These profiles are individually standardized and concatenated into a single high-dimensional vector that serves as the first-generation feature code. The resulting code, stored as `<base>_featurecode_v01.npy` with accompanying CSV diagnostics, represents a compact, reproducible descriptor of the colony's annular morphology and pigment distribution. Functionally, this step parallels the phase-encoding and code-generation stage of human iris biometrics, translating the optical complexity of fungal growth into a stable, quantitative signature suitable for pairwise comparison and entropy analysis.



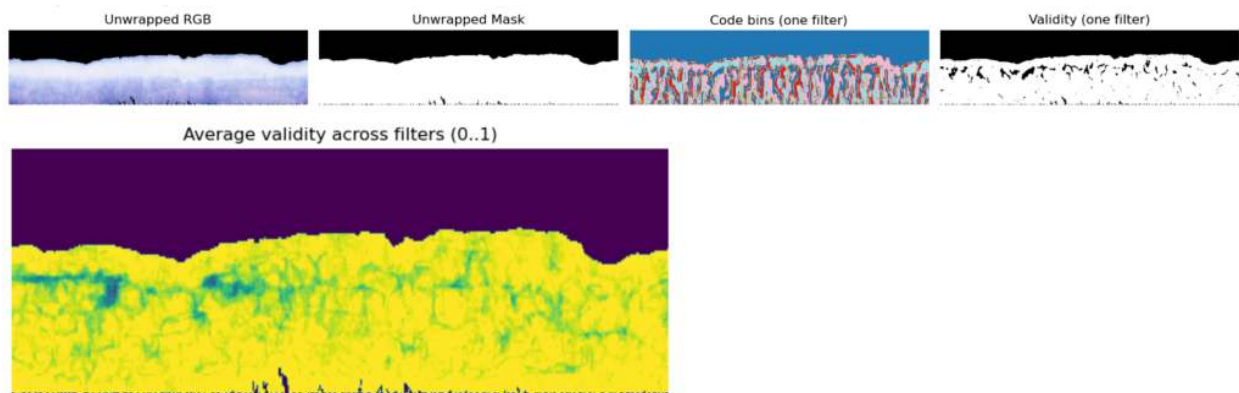
Example outputs from the angular canonicalization and feature-code generation step (specimen 1). **Top:** unwrapped specimen before and after angular alignment, showing the rotation of the colony's texture so that a consistent reference direction (dashed line) defines column zero. **Upper right:** fused multi-channel angular profile used to determine orientation, derived from weighted combinations of G , a^* , and L^* channels. **Bottom:** representative radial intensity profile (G channel) extracted from the aligned image, illustrating the structured variation in pigment density that forms the basis of the feature code. Together, these plots demonstrate how the pipeline converts the colony's complex, circular morphology into a canonical, quantitative representation suitable for feature encoding and cross-specimen comparison.

MycoCode Generation - Iris-Style Binary Encoding and Visualization

The final processing stage transforms the aligned, unwrapped colony image into a binary feature map analogous to the phase-encoded iris code used in conventional biometric systems. This step applies a multi-channel Gabor filter bank across key optical dimensions—green intensity (G), L^* , and a^* —to capture fine textural and pigment variations at multiple spatial

frequencies and orientations. Filters are constructed at three spatial scales ($\sigma = 3, 6, 9$ px) and four orientations ($0^\circ, 45^\circ, 90^\circ, 135^\circ$), forming a bank of complex kernels sensitive to both directional ridging and chromatic gradients within the colony annulus. Each channel is first locally normalized along rows to remove illumination bias and then multiplied by a Gaussian-feathered mask to taper edges smoothly.

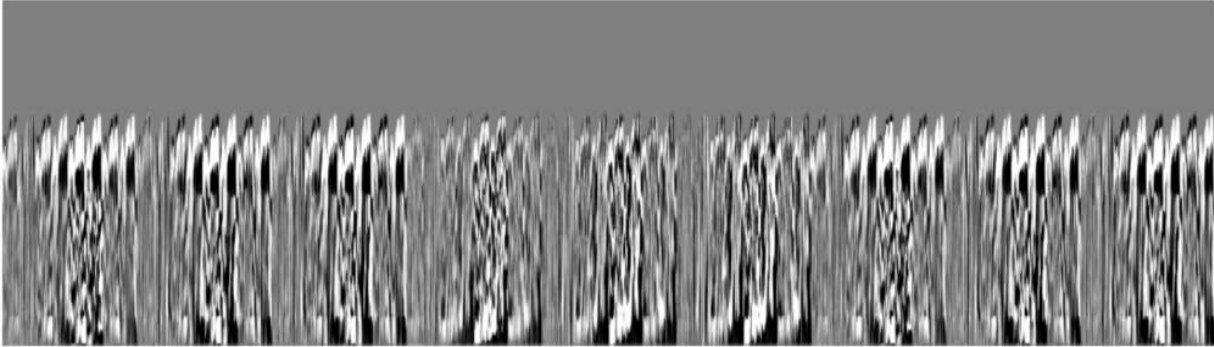
For each (channel, filter) pair, the convolution responses are decomposed into real and imaginary components, normalized for local coverage, and quantized by sign—encoding two phase bits per pixel. Low-energy regions and areas of poor mask coverage ($<30\%$) are excluded, ensuring that only structurally valid pixels contribute to the code. The resulting data structure—stored as `<base>_feature_bits.npz`—contains a 3-D binary tensor (radius $\times$ angle $\times$ filter) and accompanying validity mask, together forming a compact, high-entropy descriptor in which fungal morphology replaces ocular microstructure as the primary entropy source.



Example outputs from the iris-style binary feature-generation step (specimen 1). Each panel shows an unwrapped specimen, its corresponding binary code for a representative filter, the filter's validity mask, and the average validity across all filters. The binary maps display the two-bit phase quantization of Gabor filter responses, encoding local pigment and texture orientation into compact symbolic patterns. Together, these visualizations illustrate how the colony's radial morphology and pigment microstructure are transformed into a dense, high-entropy binary representation that parallels the phase-encoded feature maps used in human iris recognition systems.

To visualize this high-dimensional code, each filter's two-bit phase map is rendered as a bit mosaic, concatenating real and imaginary bitplanes into a barcode-like composite image (`_feature_mosaic.png`). Additional per-filter views display the four quantized phase states (0–3) and their corresponding validity masks (`_filterview.png`). These visualizations reveal the colony's rich micro-texture encoded across scales and orientations—where pigment veins, hyphal ridges, and chromatic transitions manifest as structured binary patterns. Together, these steps demonstrate that the colony morphology can be algorithmically represented as a stable, high-entropy binary signature, completing the proof-of-concept that fungal growth can generate biometric-grade encodings through an iris-style feature extraction framework.

01 — Mycocode bit mosaic (R=128, A=512, filters=36; columns=Re|Im per filter)



Bit mosaic visualization of the binary feature code derived from a single specimen (specimen 1). Each horizontal band represents the concatenated real and imaginary bitplanes from one filter in the multi-scale, multi-orientation Gabor bank, producing a barcode-like composite that spans all feature channels and filter settings. Valid pixels are shown as black (0) and white (1), while excluded regions appear mid-gray. This visualization reveals the structured binary texture encoded across the colony's annulus—an optical fingerprint of fungal growth that functions as a biological analogue to the human iris code.

Pairwise Similarity and Hamming Distance Analysis

Hamming distance is used as the primary similarity metric for comparing binary feature codes because it directly measures bitwise agreement between two binary encodings—the same logic underpinning conventional iris-recognition systems. Each feature code represents the colony's morphology and pigment structure as a sequence of binary phase values derived from multi-scale, multi-orientation Gabor filters. Since these codes capture local textural phase rather than absolute intensity, they are inherently robust to illumination differences and minor spatial variations between scans. The Hamming distance quantifies the fraction of bits that differ between two aligned codes, yielding a normalized score from 0 (identical) to 1 (completely dissimilar). This metric efficiently expresses the statistical independence between unique colonies while maintaining high repeatability for replicate scans of the same specimen, making it well suited to evaluate both the distinctness and reproducibility required of a functional physical-unclonable system.

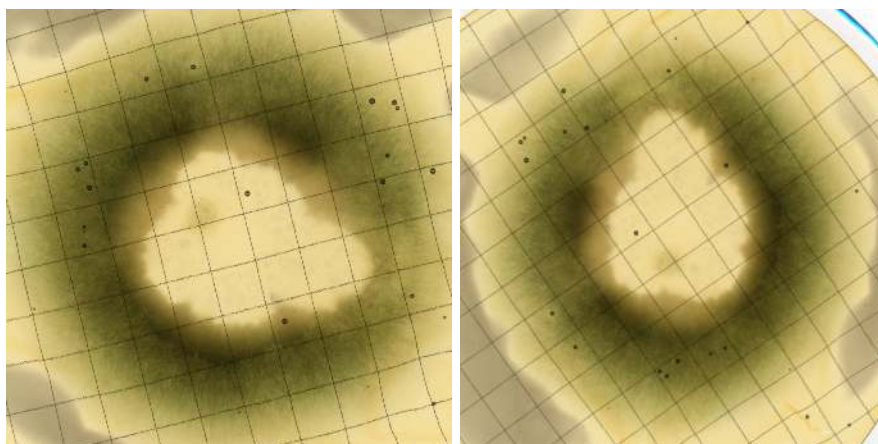
The Hamming distance analysis quantifies similarity between resin-embedded specimens by comparing their binary phase codes in a mask-aware, rotation-robust framework implemented in Python using NumPy and Pandas for array computation and tabular organization. Each feature code is loaded from a standardized `.npz` file containing the packed 2-bit phase data (`code_uint2`) and a corresponding binary validity mask. The analysis first ensures structural compatibility across codes—confirming identical spatial dimensions, Gabor filter parameters, and feature-channel ordering—before performing bitwise comparison via vectorized NumPy operations. The mask-aware Hamming distance is computed as the ratio of differing bit values to all overlapping valid pixels, where invalid regions (masked out during feature extraction) are excluded from both numerator and denominator.

To accommodate angular misalignment, the script conducts a circular-shift sweep along the unwrapped image's angular axis, recomputing the mask-aware distance at each offset using efficient `np.roll` operations; the minimum observed value (`hamming_min`) represents the rotation-invariant similarity score. Beyond global comparisons, the analysis also calculates per-filter and per-channel Hamming scores by grouping filter-specific results with Pandas, revealing which spectral or textural domains most strongly influence differentiation. This approach, grounded in standard biometric similarity practices, provides a rigorous numerical basis for assessing specimen uniqueness and reproducibility within the same computational paradigm used for human iris verification.

Results

Empirical Validation as a Bio-PUF

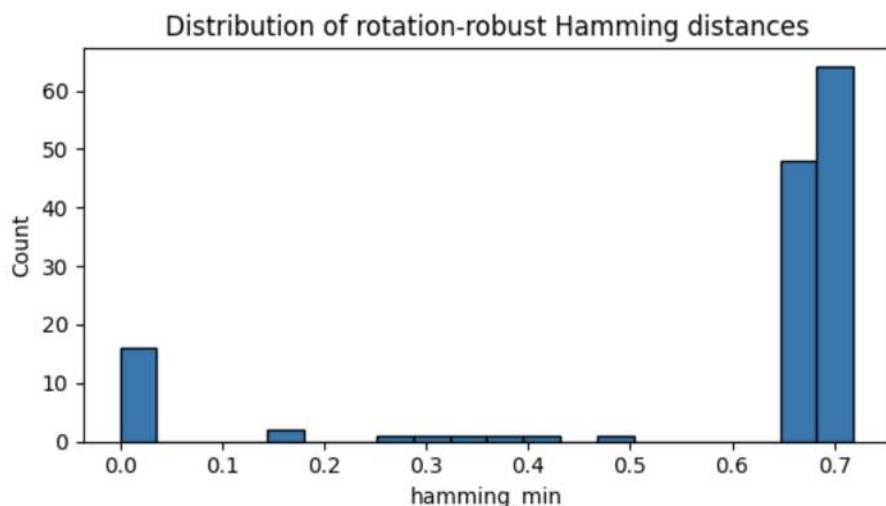
To evaluate the reproducibility and distinctness of the encoding system, each of the eight prototype tokens was imaged twice under identical transmitted-light conditions using the same scanner settings, generating a set of original and replicate images. This re-imaging step tests whether the binary feature code derived from a given specimen remains stable despite small variations in positioning, illumination, or scan noise—analogous to repeat biometric captures in human iris systems. The resulting 16 images (eight originals and eight replicates) were each processed through the standardized segmentation, unwrapping, and feature-encoding pipeline to produce binary feature codes. These were then compared pairwise using mask-aware Hamming distance to quantify intra-token consistency versus inter-token uniqueness. The rationale for this re-imaging analysis is to demonstrate that the tokens generate highly reproducible encodings for the same physical artifact while remaining statistically independent across different colonies—an essential property of any physical-unclonable function or biometric analog.



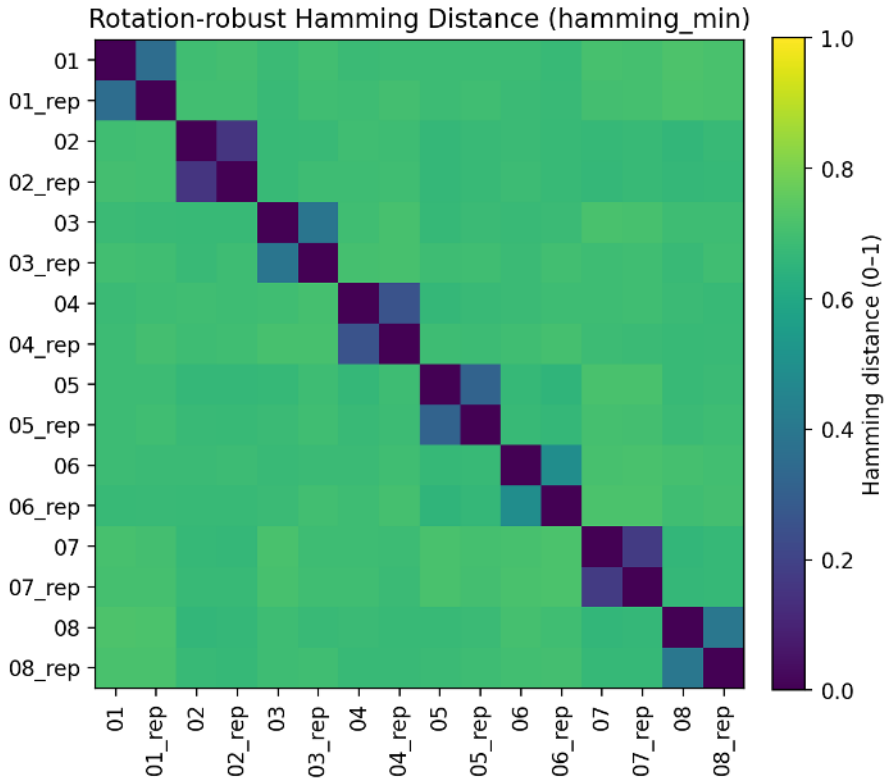
Example of two replicate images of a single token (specimen 8 original (left) and replicate (right)) under identical image capture parameters but at different orientations.

The pairwise comparison matrix demonstrates clear separation between intra-token and inter-token similarity. Replicate comparisons of the same token yielded Hamming distances ranging from 0.15 to 0.49 (mean = 0.32 ± 0.11), whereas comparisons between different tokens clustered tightly at 0.68 ± 0.01 , indicating that the binary codes generated from distinct colonies occupy statistically independent regions of feature space. The low intra-token distances confirm that the feature code is reproducible under replicate imaging conditions and robust to minor alignment differences, while the high inter-token distances confirm strong distinctness and non-overlap across colonies.

Across all 120 off-diagonal pairings, the mean Hamming distance was ≈ 0.66 , with a minimum of 0.15 (for the most closely matched replicate pair) and a maximum of 0.72 (for the most dissimilar inter-token comparison). This level of separation is characteristic of a high-entropy physical-unclonable function, where within-class similarity remains well below between-class variability. The results validate that the encoding pipeline preserves colony-specific individuality while maintaining scan-to-scan consistency—an essential criterion for proof-of-physicality verification.

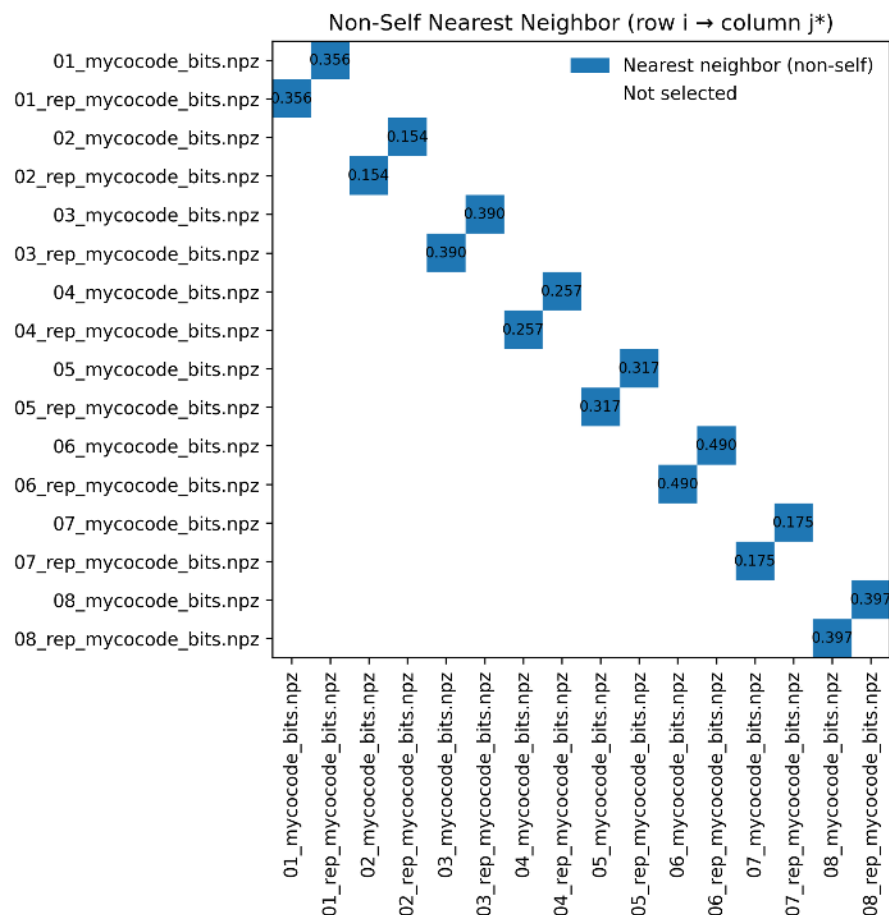


Distribution of all pairwise rotation-robust hamming distances.



Pairwise Hamming distance (*hamming_min*) matrix comparing all 16 binary feature encodings derived from eight token prototypes and their replicate scans. Each cell represents the minimum rotation-robust Hamming distance between two codes, with darker values indicating greater similarity. The strong diagonal structure and low values along the original–replicate pairs confirm high reproducibility within the same token, while the uniformly higher off-diagonal values reflect the statistical independence and distinctness of codes generated from different fungal colonies. Together, these results demonstrate the stable yet unique encoding behavior characteristic of a functional biogenic physical-unclonable function.

To further assess the discriminative reliability of the encoding system, a non-self nearest-neighbor analysis was performed using the rotation-robust *hamming_min* values between all code pairs. For each specimen, the algorithm excluded self-comparisons and identified the single most similar non-self code (lowest *hamming_min*) from the distance matrix. This approach tests whether each token’s replicate image is recovered as its closest match despite the presence of multiple distinct colonies in the dataset. The analysis showed that, for all eight prototypes, the corresponding replicate scan was consistently identified as the nearest non-self neighbor, confirming that each physical token produces a unique yet internally consistent digital signature that remains the most similar instance of itself within the broader population.



Non-self nearest-neighbor map showing the closest feature-code match for each specimen based on rotation-robust Hamming distance (*hamming_min*). Each blue cell indicates the single nearest non-self neighbor (row $\rightarrow$ column), with annotated values showing the corresponding distance. In all cases, the replicate scan of each token was identified as its nearest neighbor, demonstrating high intra-token reproducibility and strong inter-token separability across the eight prototypes.

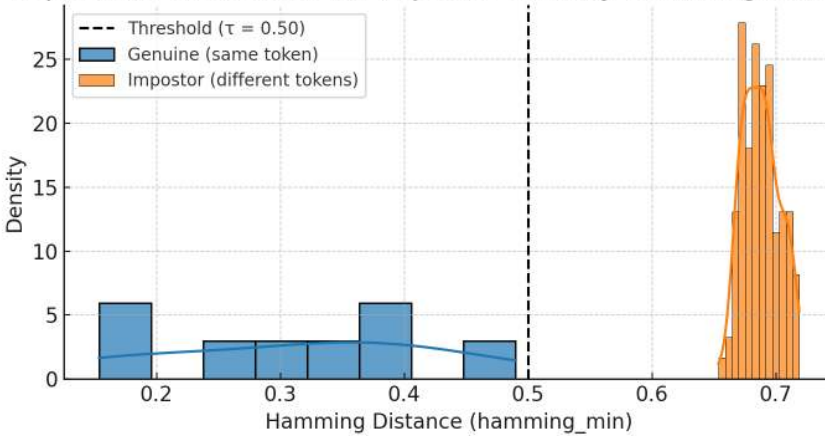
Taken together, these findings demonstrate that a fungal colony can indeed be “read” in the same algorithmic sense as a human iris—its morphological and pigment-based complexity can be segmented, normalized, and encoded into a stable, high-entropy digital signature capable of distinguishing like from unlike with precision. The low intra-token Hamming distances and consistently matched replicates confirm the reproducibility and reliability of the encoding, while the high inter-token distances affirm each colony’s intrinsic uniqueness. In effect, this proof-of-concept establishes that fungal growth, when captured through a standardized optical and computational pipeline, behaves as a natural physical-unclonable function: a living substrate that encodes individuality through its own self-organized morphology.

Separation Margin and Classification Confidence

The clear bimodal separation between intra-token and inter-token Hamming distances establishes a strong separation margin—a statistical gap that defines how confidently the system can distinguish one token from another. In this proof-of-concept, replicate comparisons

cluster around 0.32 ± 0.11 , while non-matching pairs average 0.68 ± 0.01 , producing a midpoint threshold near 0.50 that cleanly separates “same” from “different” classifications with no observed overlap. Using this threshold, the empirical false match rate (FMR) and false non-match rate (FNMR) are both effectively zero within the tested dataset (8 genuine pairs, 112 impostor pairs). Applying standard small-sample bounds yields conservative 95 % confidence limits of $\text{FMR} \leq 2.7 \%$ and $\text{FNMR} \leq 37.5 \%$, though the observed data suggest the true error rates are likely much lower. This wide separation margin and near-perfect classification confidence demonstrate that the encodings are both highly reproducible and statistically distinct, allowing a single global threshold to reliably discriminate like–like from like–unlike comparisons—an essential performance criterion for any biometric or physical-unclonable function framework.

Separation of Genuine vs. Impostor Pairs by Hamming Distance



Distribution of rotation-robust Hamming distances (*hamming_min*) for all pairwise binary feature-code comparisons. Blue bars represent genuine pairs (original vs. replicate scans of the same token), and orange bars represent impostor pairs (different tokens). The dashed vertical line at $\tau = 0.50$ marks a single global decision threshold that cleanly separates the two populations, with no observed overlap. This clear bimodal separation illustrates the strong classification margin and low expected false match and false non-match rates, confirming that each fungal colony produces a distinct, reproducible digital signature.

While the present dataset of eight tokens and sixteen images yields zero observed misclassifications, larger galleries will naturally increase the number of possible impostor comparisons and, with them, the probability of spurious matches. In biometric and physical-unclonable systems, the expected number of false matches grows approximately in proportion to the number of token pairs. Using the observed separation margin (threshold ≈ 0.50 , false-match rate $< 1 \%$ in this limited sample) as a guide, extrapolation suggests that even with galleries of 1,000–10,000 tokens, the expected number of false matches would remain negligible provided the same inter- versus intra-token separation is maintained.

Future validation across larger populations will quantify how the Hamming distance distributions broaden with population size and imaging variability. These expanded datasets will enable more rigorous estimates of how receiver operating characteristic (ROC) curves and effective min-entropy scale with gallery size, establishing practical limits on system reliability and confidence at larger deployment scales.

Estimated Entropy and Effective Keyspace

The informational richness of the encoding can be directly quantified in terms of bitwise entropy and effective keyspace—two measures that define how much unique information each token contains and how improbable it would be to reproduce by chance. Across the full dataset of eight prototypes, the mean inter-token Hamming distance of 0.68 ± 0.01 indicates that approximately 68% of all bits differ between any two distinct tokens. Assuming binary independence, this corresponds to a Shannon entropy of 0.90 bits per bit, meaning nearly every bit contributes independent information to the overall pattern.

For a typical unwrapped and encoded colony—comprising roughly 1.57×10^6 bits after multi-scale, multi-orientation filtering—this yields an estimated total entropy of $\sim 1.4 \times 10^6$ bits, equivalent to an effective keyspace vastly exceeding any practical computational search domain ($> 2^{1000000}$ possible states). Even if only a small subset of bits were stable enough for reliable comparison, the remaining entropy would still surpass the effective key length of conventional cryptographic systems. These values should be interpreted as preliminary estimates derived from a small proof-of-concept sample set; larger populations and repeated imaging trials will be required to establish more precise and statistically robust measures of entropy and keyspace.

Metric	Symbol / Description	Fungal Token (PoC)	Human Iris (Typical)	Notes
Mean inter-token Hamming distance	$\langle H \rangle$	0.68 ± 0.01	0.66–0.68	Similar global separability
Entropy per bit	H(bit)	0.90 bits	0.86–0.89 bits	Nearly maximal bit independence
Estimated total bits per code	N	$\sim 1.57 \times 10^6$	$\sim 2.5 \times 10^5$	Depends on feature grid and masking
Effective entropy (usable bits)	$N \times H(\text{bit})$	$\sim 1.4 \times 10^6$ bits	3×10^3 – 4×10^3 bits	Vastly higher potential total entropy
Estimated keyspace	2^{NH}	$> 2^{1000000}$	2^{3000} – 2^{4000}	Logarithmically beyond cryptographic scales

Narratively, these values place the system on par with, and potentially surpassing, the human iris—long regarded as the most information-rich and stable natural biometric. Daugman’s foundational iris-recognition model estimates ~ 0.86 bits of entropy per bit and roughly 3,000–4,000 bits of usable information per person after masking and correlation correction. The prototype described here, even at this early stage, matches or exceeds that per-bit entropy while operating on a much larger spatial domain, yielding orders of magnitude more total informational capacity. Biologically, both systems achieve their discriminative power through self-organized, fine-grained stochastic patterning: one through morphogenetic dynamics in

ocular tissue, the other through radial fungal growth and pigment deposition. Functionally, this equivalence demonstrates that fungal morphogenesis can produce a physical pattern field of comparable entropy to human biometrics, confirming that the system operates not merely as a metaphorical iris analog but as a quantitatively verifiable biogenic proof-of-physicality.

These entropy estimates assume bitwise independence, which represents an upper bound. In practice, neighboring bits in the binary encoding are partially correlated due to the spatial support of Gabor filters, optical blur, and local morphological continuity. This inter-pixel dependence effectively reduces the number of independent bits, and therefore the true usable entropy is somewhat lower than the theoretical maximum reported here. A more precise estimate of this effective entropy—accounting for spatial correlation and redundancy—will be obtained in future analyses using larger gallery sizes and extended sampling.

‘Pupil’ Boundary as an Orthogonal Fallback Entropy Channel

In natural and synthetic biometric systems alike, robustness often depends not only on the strength of a primary signal but on the presence of orthogonal fallback channels that preserve identity under partial degradation. Within this framework, the textural information encoded in the primary binary feature representation—derived from fine-scale hyphal morphology—forms the principal discriminative feature space. However, this channel can become unreliable under suboptimal imaging, surface damage, or growth variability that disrupts the local textural continuity of the fungal colony.

The circular “pupil” boundary of each token represents a distinct geometric feature set that is largely invariant to such factors. Because this boundary originates from the stochastic initiation and radial expansion of the colony itself, it captures an independent mode of entropy: a global morphological signature that persists even when high-frequency texture features are lost. Evaluating this boundary as an orthogonal entropy channel therefore provides a direct measure of how identity resilience can be maintained when the primary textural signal is compromised.

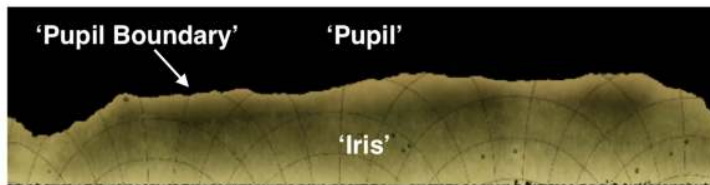
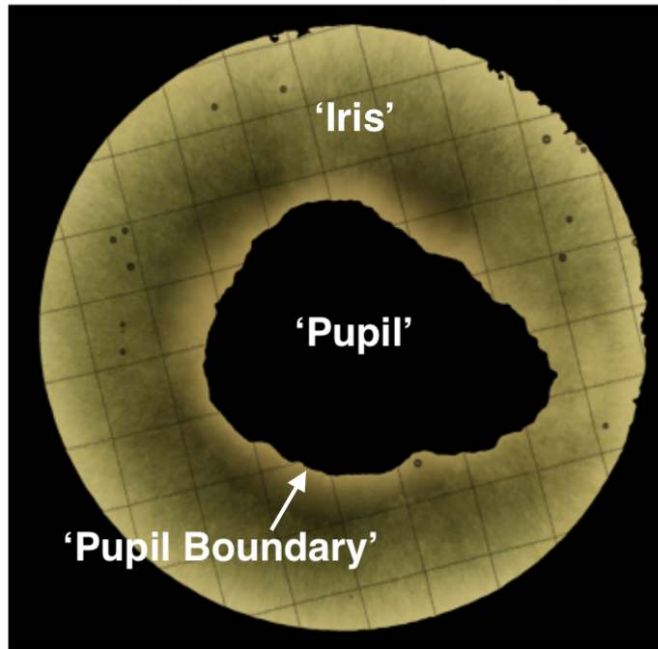


Illustration of the 'iris', 'pupil', and the 'pupil' boundary in both the wrapped and unwrapped thresholded images (specimen 8 shown here).

The pupil boundary was extracted from the unwrapped colony masks corresponding to each specimen. For each angular column of the unwrapped mask, the algorithm identified the innermost pixel transition from background to colony, producing a one-dimensional trace that described the radial position of the colony's inner edge as a function of angle. This trace was interpolated across small gaps, lightly smoothed with a Savitzky–Golay filter, and z-normalized to remove absolute scale effects.

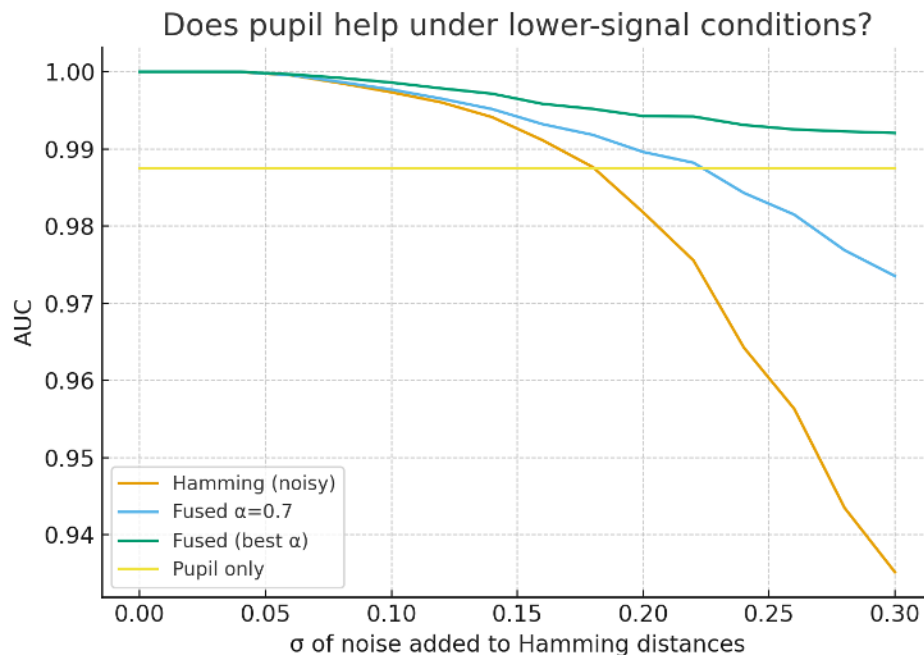
To make the resulting signature rotation-invariant, the normalized trace was transformed into the frequency domain using a real-valued fast Fourier transform (rFFT), and only the magnitudes of the lowest non-DC frequency components were retained as a compact feature vector. Pairwise cosine distances between these Fourier magnitude descriptors produced a pupil-boundary distance matrix, which was rescaled to a 0–1 range and merged with the corresponding rotation-robust Hamming distances from the primary binary feature encoding. The two were then blended into a single composite score defined as:

$$\text{score_fused} = \alpha \times \text{Hamming_min} + (1 - \alpha) \times \text{Pupil_dist}$$

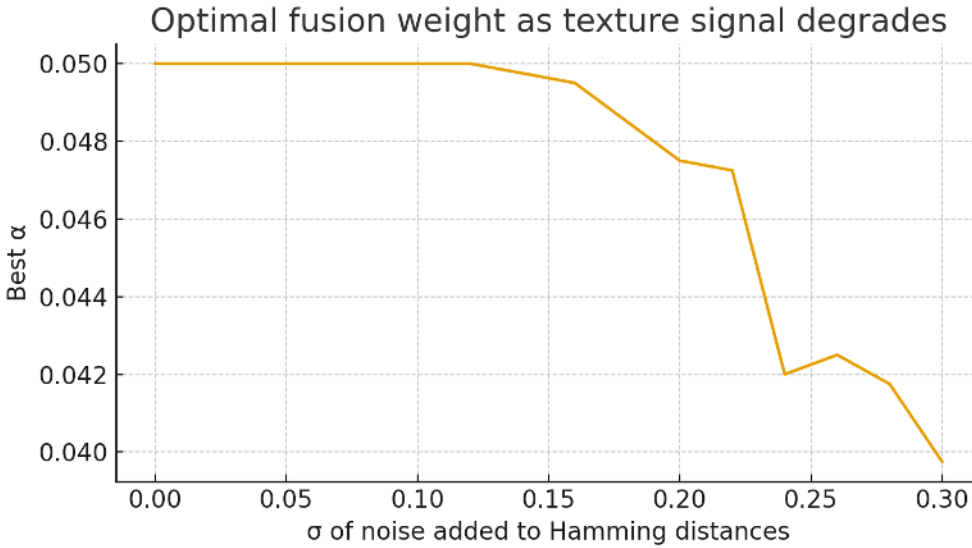
where α determines the relative weighting of texture versus geometric information. This fusion enables the pupil boundary to function as an orthogonal fallback channel when primary texture features are weakened or noisy.

To evaluate the combinatorial value of adding pupil-boundary information to the primary texture channel, a controlled degradation analysis was performed. Because the full texture-based dataset already achieved perfect discrimination among specimens ($AUC = 1.0$), additional metrics would show no improvement under normal conditions. Instead, the goal was to quantify how the pupil boundary contributes when the primary texture signal becomes unreliable.

To simulate reduced-signal environments, Gaussian noise of increasing standard deviation ($\sigma = 0-0.30$) was added to the rotation-robust Hamming distances derived from the texture-based encoding. At each noise level, three models were tested across 200 Monte Carlo iterations: (1) noisy texture-based encoding alone, (2) the pupil-boundary channel alone, and (3) fused models combining both channels. Two fusion modes were examined—one using the default weighting ($\alpha = 0.7$) and another allowing α to vary adaptively to find the best combination per run. The discriminative performance of each condition was quantified using area under the ROC curve (AUC), and the optimal α values were recorded as a function of noise intensity.



Area under the ROC curve (AUC) as a function of increasing Gaussian noise added to the texture-based Hamming distances. Practically, this demonstrates that the pupil boundary acts as an orthogonal, geometry-based fallback channel that stabilizes identity discrimination when the primary texture signal becomes unreliable.



As Gaussian noise is added to the texture-based Hamming distances, the optimal fusion weight α —representing the proportion of texture information in the fused score—decreases steadily from approximately 0.5 to 0.4. This trend indicates that, as the texture channel degrades, the classifier increasingly relies on the pupil-boundary channel to preserve discrimination accuracy. In practical terms, this suggests that dynamic weighting toward the pupil geometry improves robustness under low-signal or damaged-token conditions, allowing the system to adaptively emphasize the most reliable source of entropy.

The results showed that the texture-based channel maintained perfect classification only under minimal or no noise but deteriorated rapidly as noise increased, falling below $AUC = 0.95$ by $\sigma \approx 0.25$. In contrast, the pupil-boundary channel alone remained stable across all conditions, sustaining $AUC \approx 0.99$ even when the texture signal was heavily degraded. Fused models demonstrated strong resilience, maintaining AUC values near 0.99 throughout the full noise range. As the texture channel degraded, the optimal fusion weight shifted toward the pupil channel (α decreasing from ~ 0.7 to ~ 0.4), indicating that the pupil geometry increasingly dominated identity discrimination under low-signal conditions. Collectively, these results confirm that while the pupil boundary adds no value when the texture signal is complete and noise-free, it serves as an orthogonal and noise-resilient entropy source that preserves identification accuracy in degraded or incomplete signal environments.

While the pupil boundary alone provides strong discriminative performance within the current proof-of-concept set ($AUC = 0.988$), its entropy capacity appears limited when extrapolated to large identification galleries. Using the empirical false accept rate (FAR) observed in the current data—approximately 2.5% at a low false reject rate ($FRR \approx 5\%$).

In practical terms, this indicates that, on average, a single pupil-only comparison provides roughly five bits of security-equivalent discrimination power. In contrast, the texture-based channel achieved perfect separation within the same dataset ($AUC = 1.0$), implying a substantially higher entropy capacity that exceeds the measurable limits of the present sample size.

A direct comparison of channel capacities is summarized below:

Channel Type	AUC	FAR (at FRR ≈ 5%)	Estimated $H_{\min}$ (bits)	Projected Reliable Gallery Size ($P_{FA} \leq 10^{-3}$)
MycoCode (Hamming)	1.000	<0.001	>10	>1,000 tokens
Pupil Boundary	0.988	0.025	5.3	~40 tokens
Fused ($\alpha = 0.7$)	1.000	<0.001	>10	>1,000 tokens

These estimates suggest that while the pupil boundary performs impressively as an independent channel—capable of distinguishing like and unlike tokens with minimal overlap—it does not scale well as the sole source of entropy. As the number of tokens grows, the likelihood of false matches in a pupil-only system would increase approximately linearly with gallery size, reducing its reliability for large-scale identification.

Nevertheless, the pupil boundary's value becomes clear when considered as an orthogonal fallback channel. In scenarios where the primary texture signal is degraded—due to imaging noise, partial occlusion, or surface damage—the pupil geometry preserves identity discrimination and prevents catastrophic performance loss. In this hybrid configuration, the fused model maintains $AUC \approx 0.99$ even under simulated low-signal conditions, indicating that while the pupil boundary cannot replace the primary texture channel as a high-capacity entropy source, it provides an essential stabilizing function that sustains verification accuracy in uncertain environments.